

Comparing Methods for Multi Object Segmentation and Tracking

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Abstract

This project will compare four approaches to segmenting objects and keeping their identities consistent in video: YOLO segmentation with ByteTrack or BoT-SORT, SAM 2.1, SAM-Track, and OpenCV background subtraction. We will study people, animals, and vehicles, measuring mask quality, missed objects, identity errors, and processing cost. We will also compare pretrained models with fine-tuned versions where computing resources allow. The aim is to understand which errors each approach makes and whether adaptation helps on videos it has not seen during training.

1 Introduction

When two similar objects cross paths, a tracker may swap their identities even if their outlines remain accurate. Occlusion, changes in lighting, and camera movement can make this problem harder. We will examine how segmentation quality relates to tracking accuracy and whether fine-tuning improves both. The project covers several object categories rather than only vehicles. Our focus is maintaining object tracks; estimating whether an object is physically moving relative to the world is outside the study.

2 Methods

YOLO11s-seg will be paired separately with ByteTrack and BoT-SORT, allowing us to compare two trackers using the same segmenter (Ultralytics, 2026). SAM 2.1 will propagate prompted masks through its video memory (Meta AI, 2026). SAM-Track will use SAM on keyframes and DeAOT between them, with periodic checks for new objects (Cheng et al., 2023). For the classical baseline, we will test OpenCV’s MOG2 and KNN background subtraction methods (OpenCV, 2026), separate their foreground regions into instances, and associate these instances across frames using mask IoU.

SAM 2.1 and SAM-Track will receive box prompts and category labels from the same frozen YOLO detector. It will also help label and separate OpenCV’s foreground regions. The interval for adding new objects will be chosen on validation videos. The detector will not supply tracking IDs, and its processing time will count toward each pipeline’s runtime. Automatic runs will use no ground-truth prompts from evaluation videos. If we also test propagation from ground-truth masks, those results will be reported separately.

We plan to fine-tune YOLO’s segmenter and test the resulting weights with both trackers. We will also investigate fine-tuning SAM 2.1 and SAM-Track’s DeAOT component, keeping SAM fixed in the latter case (Yang and Yang, 2022). These two experiments depend on successful GPU and training-data pilots. To isolate weight adaptation, each comparison will keep prompts, the shared detector, tracking settings, and image preparation unchanged. OpenCV’s history, thresholds, and learning rate will be tuned on validation videos. Its online background updates and tracker settings are parameter adaptation, not neural fine-tuning.

3 Dataset and Experiments

We will select up to 200 labeled videos from the YouTube-VIS 2021 training set, covering people, dogs, cats, horses, and cars (Yang et al., 2021). Whole videos will be divided into training, validation, and evaluation sets in a 60/20/20 ratio, with category coverage in each set. Frames from one video will stay in one split. YOLO will train on sampled images with instance masks; SAM 2.1 and DeAOT need annotated video clips.

Every method will process the same frames in the same order. We will report results separately for fixed and moving cameras, since background subtraction is sensitive to camera movement. We

will also test blur, brightness changes, and JPEG compression on the same evaluation clips. These changes preserve the mask coordinates. Model selection and parameter tuning will use only training and validation data. We will record annotation coverage, any known overlap with model pretraining, and the splits, checkpoints, and settings used in each experiment.

4 Evaluation

We will report the three original MOTS measures (Voigtlaender et al., 2019). Multi-Object Tracking and Segmentation Accuracy (MOTSA) counts correct matches and penalizes false positives and identity switches. Soft Multi-Object Tracking and Segmentation Accuracy (sMOTSA) weights each correct match by mask intersection over union (IoU), adding sensitivity to segmentation quality. Both scores are normalized by the total ground-truth instance masks across the evaluated frames. Multi-Object Tracking and Segmentation Precision (MOTSP) is the mean mask IoU of matched instances; it excludes missed objects.

We will supplement these with mask-based Higher Order Tracking Accuracy (HOTA), its Detection Accuracy (DetA) and Association Accuracy (AssA) components, and the companion Localization Accuracy (LocA) measure (Luiten et al., 2020). HOTA balances detection and identity association; LocA measures overlap for matched masks. We will also report the identity F1 score (IDF1), identity switches (IDS), false positives (FP), and false negatives (FN), using mask-based matching. Higher accuracy, overlap, and IDF1 scores are better; lower error counts are better. sMOTSA and HOTA will be our main comparison measures.

MOTS matching requires non-overlapping masks and mask IoU above 0.5; HOTA uses its standard threshold sweep. We will check category mapping, frame indices, absent objects, ignore regions, and overlap resolution before computing scores. The five-category conversion needs validation rather than an unchanged car/pedestrian-only loader. We will describe these as MOTS-style results on our YouTube-VIS subset, with native video AP as an additional measure.

For robustness, we will measure changes from each clean clip to its corrupted versions and inspect identity failures around occlusion. Speed, peak GPU memory, and training time will show

the cost of each method. Timing will include detection, prompting, object discovery, and postprocessing. We will estimate uncertainty by resampling videos where practical. Mask overlap between adjacent frames alone cannot measure stability, because the objects themselves move.

5 Computing Resources

Our lab GPU allocation is still unconfirmed. We will ask the instructor about the available GPU models, memory, access limits, and computing budget before committing to SAM 2.1 or DeAOT fine-tuning. A pilot of 100 inference frames and 20 training steps per neural candidate will check memory, gradients, checkpoint loading, and expected training time. We will start with smaller checkpoints and record which components are trainable. If training does not fit, we will report pretrained results where inference is feasible and explain the missing experiment.

References

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